Questions by topic: Atmospheric pollution: smoke, acid rain, NOx, hydrocarbons, CO, smogs - Set 2 (15 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the annual mean concentrations of four pollutants measured at a city-centre monitoring station, with the legal limit for each.

Table 1

Pollutant	Annual mean / µg m^{-3}	Legal limit / µg m^{-3}
Nitrogen dioxide	52	40
PM10 particulates	28	40
PM2.5 particulates	19	20
Sulfur dioxide	8	125

0 1 . 1 Calculate by what percentage the annual mean nitrogen dioxide concentration exceeds the legal limit.

[2 marks]

Answer ____ %

0 1 . 2 Suggest why the sulfur dioxide concentration is now far below its limit while nitrogen dioxide is not.

[3 marks]

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0 1 . 3 Explain how each of the following reduces emissions.

[3 marks]

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Electrostatic precipitator .....

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Catalytic converter .....

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Low-NOx burner .....

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0 1 . 4 Explain why a reduction in the city's NOx emissions would also reduce summer ground-level ozone.

[3 marks]

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0 1 . 5 The council also plans to monitor air quality using lichens. Explain the advantages and the limitations of using lichens as indicator species.

[4 marks]

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END OF QUESTIONS

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